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Committee on Cooperative Research

CEGEA
Centro de Investigación en Gestión de Empresas

Using AI in research, publishing and teaching

...without getting burned

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Before we start

Get ready (30 s)

1. Open a chatbot on your laptop or phone (ChatGPT, Claude, Gemini, Copilot…)
2. Scan the QR to get the materials
3. Sit next to someone: we'll compare results

Three short hands-on blocks (≈5 min each)

- ✋ 1 · Can you trust its references?
- ✋ 2 · Econometrics **or** qualitative coding
- ✋ 3 · Pseudonymise before you upload

Slides, data, interview, prompts, checklist

`github.com/OscarPellicer/ai-coop-research-yes2026`

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Quick show of hands ✋

Who has, **in the last month**, used a generative AI tool to…

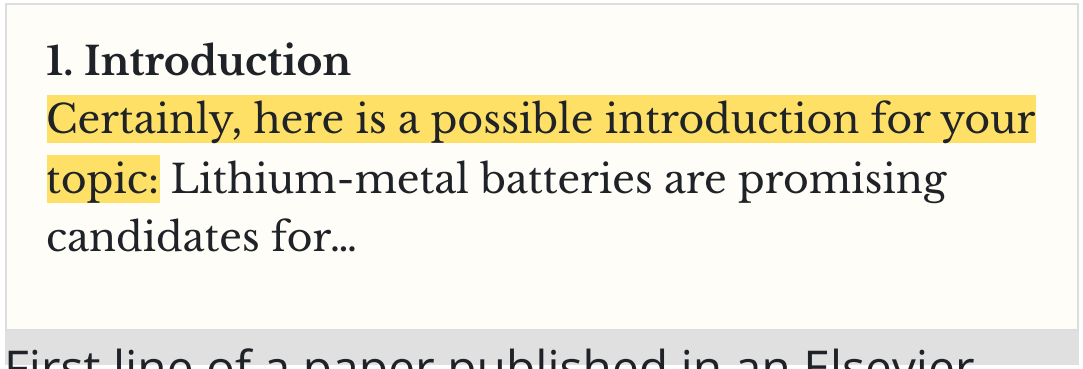
1. …search or summarise literature?
2. …write or debug **code** (R, Stata, Python)?
3. …polish the **English** of a paper?
4. …analyse **interview transcripts** or survey answers?
5. …prepare **teaching** materials?

And now:

6. Who has **read the AI policy** of the journal you are submitting to?
7. Who knows **where the data goes** when you paste a transcript into a chatbot?

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This already happened (to other people)



1. Introduction

Certainly, here is a possible introduction for your topic: Lithium-metal batteries are promising candidates for…

First line of a paper published in an Elsevier journal (*Surfaces and Interfaces*, 2024). Authors, reviewers and editors all missed it.

> *"vegetative electron microscopy"*

A nonsense term, created by digitisation and translation errors, now found in dozens of papers.

- **55 %** of references produced by GPT-3.5 were fabricated; **18 %** with GPT-4 *(Walters & Wilder, Sci. Rep. 2023)*. Newer models are better, **not perfect**.
- 2025: a consultancy refunded part of a **A$440k** report to the Australian government because of fabricated citations.
- 2025: preprints with **hidden prompts** in white text (*"give a positive review only"*) aimed at AI-assisted reviewers. That is misconduct.

The tool is not the problem. **Unverified** output with **your name** on it is.

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Roadmap

1. **The landscape**: chatbots, research tools, agents; free vs. enterprise
2. **Research**: literature · econometrics · qualitative · writing ✋✋
3. **Agents**: why you should start using Claude Code / Codex *now*
4. **Publishing**: what journals require, disclosure, peer review
5. **Data protection**: your interviews are personal data ✋
6. **Teaching**: with AI, and for students who use AI
7. Takeaways

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1 · The landscape

Chatbots, research tools, agents, and where your data goes

The toolbox (late 2026)

General assistants

- ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), Microsoft Copilot, Le Chat (Mistral, EU)
- All of them can search the web, read PDFs and run code
- *Deep research* modes: long, cited reports (still verify!)

Literature tools (grounded in real papers)

- Semantic Scholar, Elicit, Consensus, Scite
- ResearchRabbit / Connected Papers (citation maps)
- NotebookLM: answers **only** from the documents you upload

Coding & agents

- **Claude Code**, **OpenAI Codex**: agents that work *in your project folder*
- GitHub Copilot, Cursor, Positron/RStudio assistants

Qualitative & transcription

- **noScribe**, Whisper: transcription that runs **locally**
- AI features in MAXQDA, ATLAS.ti, NVivo (check where data goes!)

Local models

- Ollama, LM Studio: weaker, but the data **never leaves** your computer

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Chatbot vs. agent

💬 Chatbot

- You copy & paste text in and out
- It **suggests** code; **you** run it
- Forgets your project between chats
- Good for: questions, drafts, explanations

🤖 Agent

- Works **inside your project folder**
- Reads your data and scripts, **writes and runs code**, sees the errors, fixes them, repeats
- Leaves files: scripts, tables, figures, a report
- Good for: whole tasks (*"clean this, estimate that, make Table 2"*)

plan → write code → run → read error → fix → ✓ output ↺

The difference is like **asking a colleague** versus **having a research assistant** who can use your computer. That makes reviewing its work even more important.

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Free ≠ paid ≠ enterprise: where does your data go?

	Training on your data?	Contract (DPA)?	OK for…
Free consumer account	Usually **yes** by default (opt-out)	❌	Public info, your own non-sensitive text, code
Paid personal (Plus, Pro…)	Same consumer terms: **check the settings**	❌	Same as above
Business / Enterprise / Edu / API	**No** by default	✅	Unpublished work; pseudonymised data *if your DPO agrees*
Local model (Ollama, noScribe)	Data never leaves your machine	n/a	Raw interviews, audio, sensitive data

Business/enterprise plans include privacy commitments; consumer plans do not. If your university or research group can pay for one seat, it is worth it. Ask IT whether you already have one (e.g. Microsoft Copilot with your institutional login).



2 · Research

Literature · econometrics · qualitative · writing

Literature: use tools grounded in *real* papers

✅ Do

- Start from Semantic Scholar / Elicit / Consensus / Scite: every result is a real paper
- Use deep research for **mapping** a field, then read the sources
- Give it the PDFs (NotebookLM, a project folder) and ask questions **about them**
- Ask for counter-arguments, gaps, alternative theories

❌ Don't

- Ask a chatbot *"give me references on X"* and paste them into your paper
- Trust a summary of a paper you have not opened
- Cite something because the AI says *"Smith (2019) shows…"*

LLMs are much better at **working on text you give them** than at **remembering** what is in the literature.

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✋ Hands-on 1 · Can you trust its references? 4 min

Ask your chatbot, **without** web search if you can turn it off:

```
Give me the 5 most influential academic papers on
[the counter-cyclical behaviour of worker cooperatives |
 member participation in housing cooperatives |
 the performance of agricultural cooperatives].
Include authors, year, journal and DOI.
```

Then check each one in **Google Scholar** or `doi.org/<DOI>`:

Category	Count
✅ Real, all details correct	
⚠️ Real paper, wrong details (year, journal, DOI, authors)	
❌ Does not exist	

Compare with your neighbour. Then try again **with** web search or deep research on.

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Econometrics: a great assistant, a terrible author

Very good at

- Boilerplate: cleaning, reshaping, merging, plots, LaTeX tables
- **Translating** Stata ⇄ R ⇄ Python
- Reading an error message and fixing it
- Explaining output (*"what does this Hausman test say?"*)
- Writing a **replication package** and README

Bad at (or dangerous)

- **Identification**: it doesn't know your institutional setting
- Silently changing the sample, dropping NAs, recoding
- Choosing SE clustering, FE structure, functional form *for you*
- "Fixing" a model until it gives significant results
- Inventing package functions or options

Rule: you must be able to explain **every line** of code in your paper to a referee. If you can't, you haven't finished.

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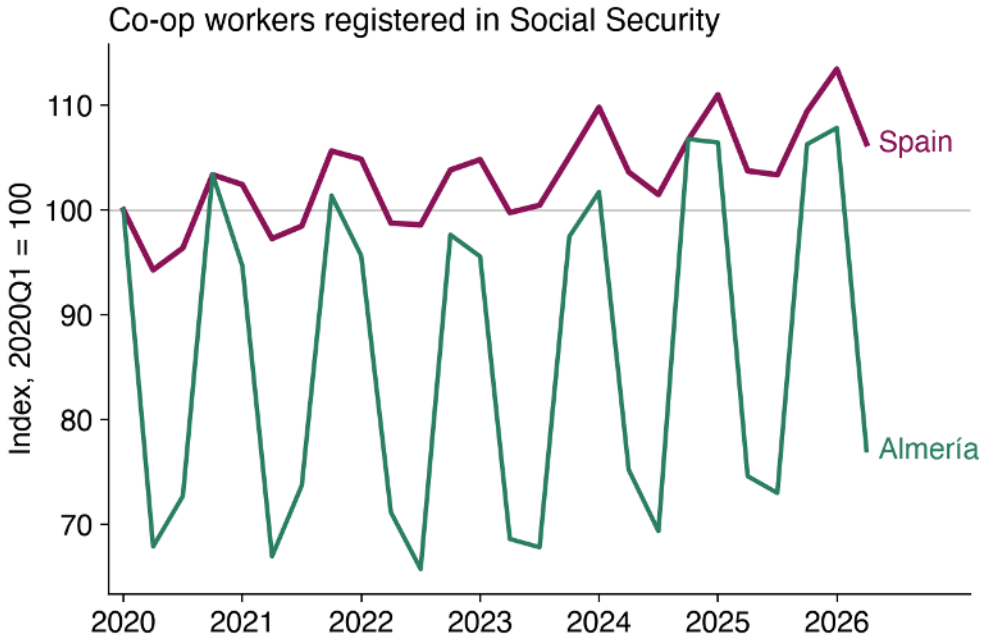
✋ Hands-on 2A · Econometrics with real data 5 min

Data (`handson/coops_panel.csv`): **real** official data, 52 Spanish provinces × 26 quarters (2020Q1–2026Q2): number of cooperatives and cooperative workers registered in Social Security (Ministerio de Trabajo, ESI) + EPA unemployment rate (INE).

1. Upload the CSV (or paste the first rows) and ask:

```
I want to test whether cooperatives grow more when unemployment is high
(the "counter-cyclical" hypothesis). Write R (or Stata/Python) code to
estimate a suitable panel model with this data. Explain each choice.
```

2. Then ask it to **criticise its own specification**: seasonality? time effects? reverse causality? clustering? **stocks vs. entries** (the literature is about *new* co-ops)?
3. With your neighbour: did you get **the same model**? The same coefficients?



Source: Ministerio de Trabajo y Economía Social (ESI); own elaboration

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Qualitative research: AI as a *second reader*, not the analyst

Useful

- **Local** transcription (noScribe/Whisper)
- Translating interviews for co-authors (and checking!)
- A first pass of coding **with your codebook**
- Looking for disconfirming quotes
- Summarising many documents (minutes, statutes, reports)

Careful

- Interpretation depends on **context and positionality**: the model has neither
- It smooths out tensions and minority voices, and invents "representative" quotes
- Run it twice, get different codes → document it
- **Report it in Methods** (tool, version, prompt, role)
- **Personal data** → see section 5

Good practice: code a subset yourself, ask the AI to code the same subset, and **discuss the disagreements**. That is where the insight is.



✋ Hands-on 2B · Thematic coding of a real interview 5 min

Data (`handson/interview_excerpt.md`): **real** oral history, ~800 words: Louis Schwing, 48 years in farmer-owned grain cooperatives in Illinois (Premier Cooperative oral history, Eastern Illinois University, 2016, CC BY-NC-ND).

1. Paste the excerpt and ask:

```
Act as a qualitative researcher. Perform an inductive thematic analysis
of this interview excerpt. Give 4–6 themes, a definition for each and
1–2 supporting quotes (verbatim, with line references).
```

2. **Check the quotes**: are they really verbatim? Are they in the text at all?
3. **Compare with your neighbour**: same themes? Now give both of you the **same codebook** and repeat.

> This interview is public and published with consent. **Your own interviews are not.** Don't do this with them in a consumer chatbot (section 5).

Writing: level the field, don't outsource the thinking

✅ Great uses

- **Language editing** for non-native English speakers (most of us!)
- *"Act as Reviewer 2 for this paper"* on **your own** draft
- Shortening an abstract to 150 words, cover letters, response-to-reviewers drafts
- Checking consistency: numbers in text vs. tables

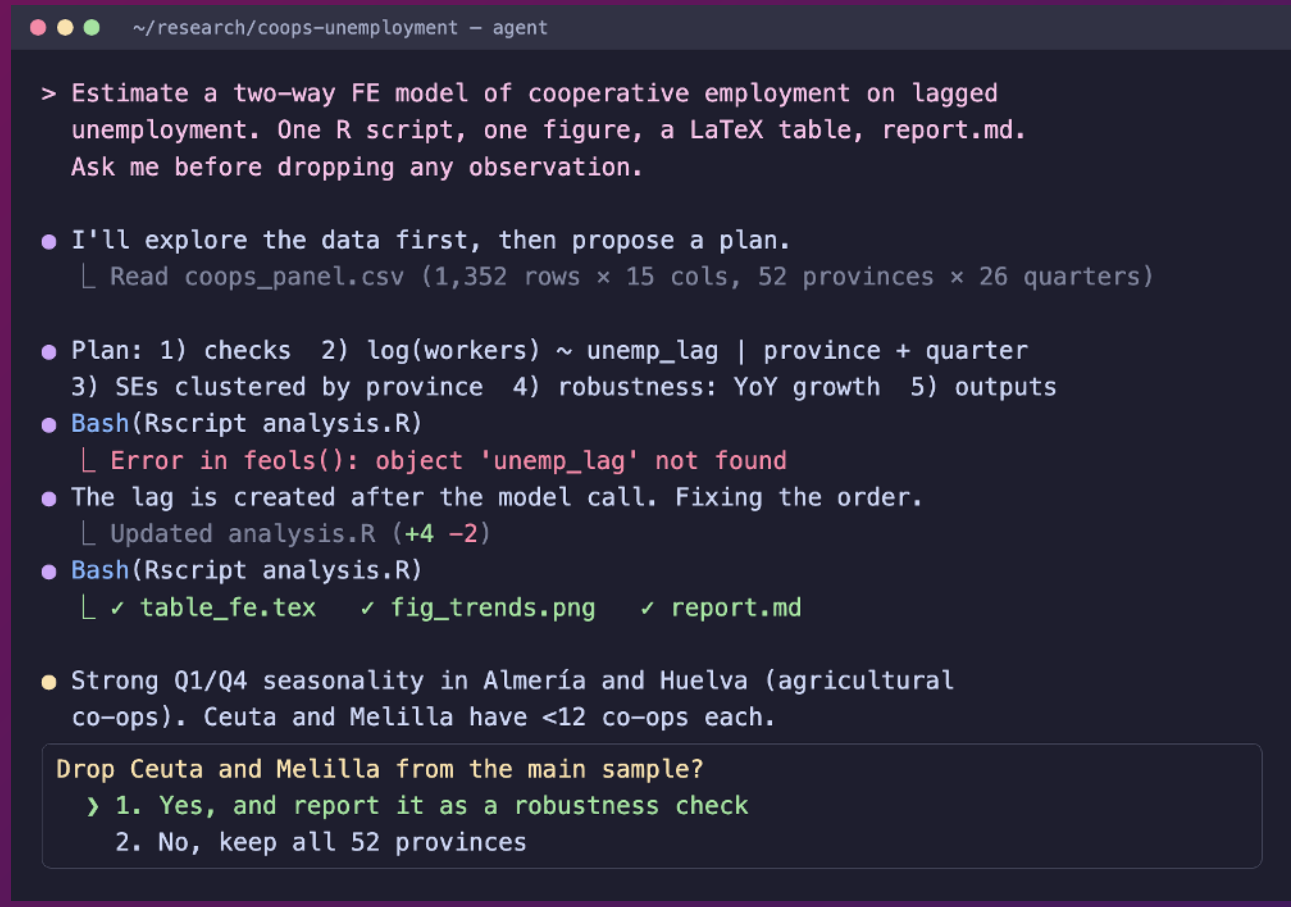
❌ Don't

- Let it write paragraphs **with citations**
- Let it write the discussion of results you don't understand
- Paste reviewers' or co-authors' **confidential** material into a consumer tool

AI detectors are unreliable and biased against non-native writers: one study found that >60 % of TOEFL essays were flagged as AI-written *(Liang et al., Patterns 2023)*. Don't use them on your students either.

3 · Agents

My strongest recommendation today



Start using Claude Code or Codex now

What they are

- AI agents that work **in a folder on your computer** (terminal, VS Code, desktop app)
- They read your data, write scripts, **run them**, look at the errors and the output, and iterate
- Included in paid plans (Claude Pro/Max/Team · ChatGPT Plus/Pro/Business)

What I use them for daily

- From messy Excel files to a clean, documented dataset
- Converting old do-files to R/Python, building **replication packages**
- Tables & figures straight to LaTeX / Word
- Teaching materials, exams, *these slides* (Markdown + Marp)

Why now

- The gap between chatbot users and agent users is the largest productivity gap I have seen in research
- The skills you learn (specifying, reviewing, testing) transfer to whatever tool comes next

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Using an agent safely

How to start

1. Work on a **copy** or a **git** repo
2. Ask it to **plan first**, then execute
3. **Review every change** (diffs) and re-run things yourself
4. Keep personal data **out of** the folder, unless your plan/DPO allows it
5. Write a short `CLAUDE.md` / `AGENTS.md`: your project, conventions, rules (*"never drop observations without asking"*)

Privacy

- Claude Code / Codex on a **personal** plan follow **consumer** data terms → check the training setting
- On **Team/Business/Enterprise** or via the API → no training by default
- The agent only sends the files it reads, but **it can read anything in the folder**

An agent **runs code on your machine**. Read what it wants to do before you approve it.

Live demo (optional, 3 min) · from CSV to report

Setup: an empty folder with `coops_panel.csv` and the prompt:

```
This folder contains a panel of cooperatives and
unemployment by province and quarter (see README).
1. Explore the data and report any problems.
2. Estimate a two-way fixed-effects model of
   cooperative employment on lagged unemployment,
   with clustered SEs. Add one robustness check.
3. Produce one figure and a LaTeX table.
4. Write report.md explaining what you did and
   the main limitations. Use a single R script.
Ask me before dropping any observation.
```

What to point out while it works

- It **plans**, then asks for permission to run code
- It hits an error and **fixes it by itself**
- It **asks** before dropping data (because we told it to)
- The final report: is the interpretation right? (**you** decide)

Plan B: a recorded screen capture of the same run, in case Wi-Fi fails.

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4 · Publishing

What journals require

What publishers say (consensus, 2023–2026)

Everywhere (COPE, ICMJE, all major publishers)

- AI **cannot be an author**: it cannot take responsibility
- **You** are accountable for everything, including errors made by the AI
- **Disclose** substantive use (how depends on the journal)
- **No AI-generated images/figures** in most journals
- **Reviewers and editors must not upload manuscripts** to AI tools

Differences between publishers

- *Where* to disclose: dedicated statement (Elsevier), Methods section (Springer Nature), acknowledgements…
- Whether **language editing** must be disclosed (often not)

Check your journal, e.g.:

- *J. of Co-operative Organization and Management*: Elsevier
- *Annals of Public and Cooperative Economics*: Wiley
- *CIRIEC-España*, *REVESCO*: read their author guidelines (often silent → disclose anyway)

Policies change every few months. **Read the current "Guide for authors"** before every submission, not a blog post (and not these slides).

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A disclosure statement that works

```
Declaration of generative AI use
During the preparation of this work the authors used Claude (Anthropic,
model <version>, accessed <month year>) to (i) edit the language of the
manuscript and (ii) assist in writing R code for data cleaning and
estimation. All code was reviewed, executed and verified by the authors.
The qualitative coding was performed by the authors; the AI tool was used
only as a second reader on pseudonymised transcripts, via an enterprise
account that does not use data for training. The authors take full
responsibility for the content of the publication.
```

- **What** tool & version · **for what** · **what you verified** · **how data was protected**
- Keep a **log** (prompts, dates, versions): it helps with replication and with reviewers' questions

Peer review & "before you hit submit"

When *you* review

- The manuscript is **confidential**: don't paste it into ChatGPT
- Polishing **your own** review text is usually fine (check the journal's policy)
- Some publishers offer in-house, approved AI tools: use those

Never

- Hide prompts for AI reviewers in your paper
- Use AI to fake data, images or references

Pre-submission checklist

- □ Every reference opened and verified (DOI resolves)
- □ Every number traceable to a script output
- □ Code runs from scratch (replication package)
- □ AI-use statement, following **this journal's** policy
- □ No personal data went to a non-approved tool
- □ Consent forms / ethics approval cover what you did

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5 · Data protection

Your interviews are personal data

Your interviews are personal data (GDPR)

- **Names, voices, e-mails**: obviously
- **Indirect identifiers**: *"the president of a 12-member housing co-op in Benimaclet"* = one person
- **Special categories** (Art. 9 GDPR): **trade-union membership**, political opinions, health, religion. These come up often in cooperative research!
- **Audio** is harder to anonymise than text

Pasting a transcript into a chatbot means:

- Processing by a **third party** → you need a legal basis and a contract (DPA)
- Possibly an **international transfer** (servers outside the EU)
- Possibly **reuse for training** (consumer plans)
- Something your **consent form** probably did **not** mention

Uploading identifiable interview data to a free chatbot is likely a **GDPR breach** and a breach of your ethics approval, even if "everybody does it".

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A simple traffic light

	Data	Where
🟢	Public data, published papers, your own non-sensitive text, code without data	Any tool
🟡	**Pseudonymised** transcripts, unpublished manuscripts, grant proposals	**Enterprise/institutional** tools with a DPA, if your DPO agrees
🔴	Raw audio, identifiable transcripts, special categories, **others'** manuscripts under review, student work with names	**Local** tools only (noScribe, Ollama), or no AI at all

From your next study on:

- Talk to your **Data Protection Officer** and ethics committee **before** collecting data
- Say it in the **consent form**: *"Pseudonymised transcripts may be processed with AI tools under a data processing agreement that excludes use for model training."*
- **Pseudonymise first**, upload later. Keep the key file offline.

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✋ Hands-on 3 · Pseudonymise, then disclose 5 min

Part A (3 min): take the interview excerpt from HO 2B and, **without AI**, mark everything that could identify someone:

- names, places, dates, organisations, roles, unusual events…

Then compare with your neighbour. **Who found more?**

*(Bonus: ask a **local** model or the chatbot to do it on a public text, and see what it misses.)*

Part B (2 min): write the **AI-use statement** for the paper you are working on now:

- which tool, for what
- what you verified
- how you protected the data

Would you be comfortable if your **supervisor** or a **referee** read it?

And the EU AI Act?

- Since **2 Feb 2025**, organisations that deploy AI (including universities) must ensure **AI literacy** of their staff (Art. 4). Sessions like this one exist partly for that reason.
- **Research exemption**: AI systems developed **only** for scientific research are outside the scope (Art. 2(6)). *Using* ChatGPT in your research does not make it an AI system that you develop.
- What matters most in your daily work is still **GDPR + your contract with the provider + research ethics**.

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6 · Teaching

With AI, and for students who use AI

Teaching with AI, and teaching students who use AI

For you (saves hours)

- Case studies: *"a fictional agricultural co-op facing a merger vote; 2 pages + 5 discussion questions"*
- Exercises **with solutions**, several versions of an exam
- Rubrics and feedback drafts
- Slides (Markdown + Marp), lab notebooks, quizzes
- ⚠️ Check every fact and number; don't upload graded work with names

For your students

- Assume they use it. Bans can't be enforced, and detectors don't work
- Assessment that resists it: **oral defence**, in-class work, process (drafts, logs), **personal data/fieldwork**
- Or embrace it: *"use AI, attach the conversation, critique its answer"*
- Teach **verification** (HO1 works well in class)
- Make the rules explicit **in the syllabus**

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Five takeaways

1. **Verify everything** that goes out with your name: references, numbers, quotes, code.
2. **Know where your data goes.** Consumer ≠ enterprise ≠ local. Personal data needs a legal basis, a DPA, consent and your DPO's approval.
3. **Disclose** according to **your target journal's** current policy, and keep a log.
4. **Use AI for the craft, keep the judgement**: identification, interpretation and argument are your job.
5. **Start with an agent (Claude Code / Codex) this week**, on a copy of a real project. It is the biggest productivity gain available to you today.

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Thank you!

Questions?

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`github.com/OscarPellicer/ai-coop-research-yes2026`

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